FDA Submission

Your Name: Sabrina Palis

Name of your Device: PneumoAssist AI

Algorithm Description

1. General Information

Intended Use Statement:
This algorithm is intended to assist radiologists in identifying the presence or absence of pneumonia on frontal-view chest X-ray images. It is not intended to provide a definitive diagnosis or to replace clinical judgment.

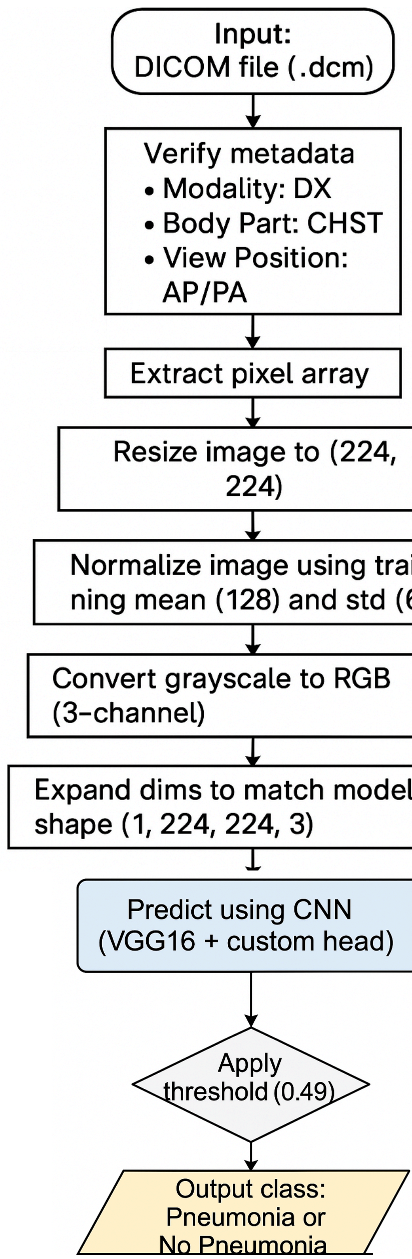
Indications for Use:
The device is indicated for use in adult patients aged 18 years and older undergoing routine screening or diagnostic chest radiography. It is intended to be used on frontal-view chest X-rays acquired using standard radiographic techniques. The algorithm is not indicated for use in pediatric patients or patients imaged in non-frontal positions or with thoracic hardware.

Device Limitations:
The model was trained exclusively on frontal-view (PA or AP) chest X-rays from the NIH dataset. These images were labeled using NLP on radiology reports, which may include inaccuracies. The model has not been validated on lateral views, pediatric cases, or non-standard imaging such as CT. Comorbidities like effusion or fibrosis may reduce performance.

Clinical Impact of Performance:
False negatives may delay treatment, while false positives may result in unnecessary follow-up. The algorithm is intended solely as a decision support tool. All clinical decisions must remain with the physician.

2. Algorithm Design and Function

Algorithm Flowchart:



DICOM Checking Steps:

- Each .dcm file is loaded using pydicom.
- Metadata fields are validated to ensure compatibility:
- Modality must be 'DX'
- View Position must be 'AP' or 'PA'
- Body Part Examined must be 'CHEST'
- Files not matching these criteria are rejected before inference.
- Pixel data is extracted only if available and readable.

Preprocessing Steps:

- Image resized to 224×224 pixels using skimage.transform.resize.
- Pixel values normalized using training mean 128.0 and standard deviation 64.0:
(img−128.0)/64.0(img−128.0)/64.0
- Grayscale image expanded to 3 channels by duplication.
- Image reshaped to match model input: (1, 224, 224, 3).

CNN Architecture:

The model architecture is based on the VGG16 convolutional neural network pretrained on ImageNet. All convolutional layers (totaling over 14 million parameters) were initially frozen to retain their learned feature extraction capabilities. On top of the VGG16 base, several custom layers were added to adapt the model to the binary classification task of detecting pneumonia. These include a `GlobalAveragePooling2D` layer to reduce spatial dimensions, followed by two `Dropout` layers (to mitigate overfitting), a dense layer with 128 ReLU-activated units, and a final sigmoid-activated dense output layer. Only the custom top layers were made trainable during fine-tuning, resulting in approximately 66,000 trainable parameters.

```
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
input_2 (InputLayer)         [(None, 224, 224, 3)]     0
_________________________________________________________________
block1_conv1 (Conv2D)        (None, 224, 224, 64)      1792
_________________________________________________________________
block1_conv2 (Conv2D)        (None, 224, 224, 64)      36928
_________________________________________________________________
block1_pool (MaxPooling2D)   (None, 112, 112, 64)      0
_________________________________________________________________
block2_conv1 (Conv2D)        (None, 112, 112, 128)     73856
_________________________________________________________________
block2_conv2 (Conv2D)        (None, 112, 112, 128)     147584
_________________________________________________________________
block2_pool (MaxPooling2D)   (None, 56, 56, 128)       0
_________________________________________________________________
block3_conv1 (Conv2D)        (None, 56, 56, 256)       295168
_________________________________________________________________
block3_conv2 (Conv2D)        (None, 56, 56, 256)       590080
_________________________________________________________________
block3_conv3 (Conv2D)        (None, 56, 56, 256)       590080
_________________________________________________________________
block3_pool (MaxPooling2D)   (None, 28, 28, 256)       0
_________________________________________________________________
block4_conv1 (Conv2D)        (None, 28, 28, 512)       1180160
_________________________________________________________________
block4_conv2 (Conv2D)        (None, 28, 28, 512)       2359808
_________________________________________________________________
block4_conv3 (Conv2D)        (None, 28, 28, 512)       2359808
_________________________________________________________________
block4_pool (MaxPooling2D)   (None, 14, 14, 512)       0
_________________________________________________________________
block5_conv1 (Conv2D)        (None, 14, 14, 512)       2359808
_________________________________________________________________
block5_conv2 (Conv2D)        (None, 14, 14, 512)       2359808
```

```
_________________________________________________________________
block5_conv3 (Conv2D)        (None, 14, 14, 512)       2359808
_________________________________________________________________
block5_pool (MaxPooling2D)   (None, 7, 7, 512)         0
_________________________________________________________________
global_average_pooling2d_1 ( (None, 512)               0
_________________________________________________________________
dropout_2 (Dropout)          (None, 512)               0
_________________________________________________________________
dense_2 (Dense)              (None, 128)               65664
_________________________________________________________________
dropout_3 (Dropout)          (None, 128)               0
_________________________________________________________________
dense_3 (Dense)              (None, 1)                 129
=================================================================
Total params: 14,780,481
Trainable params: 65,793
Non-trainable params: 14,714,688
```

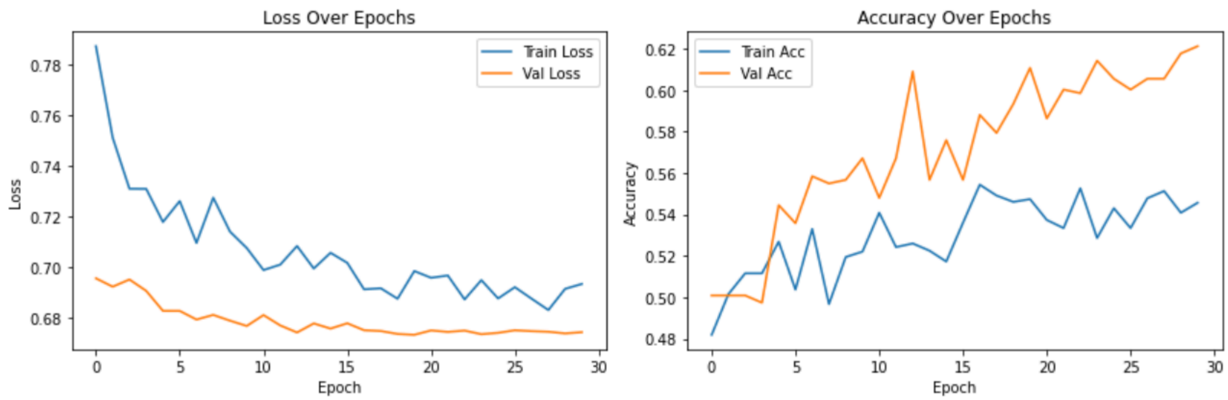
3. Algorithm Training

Parameters:

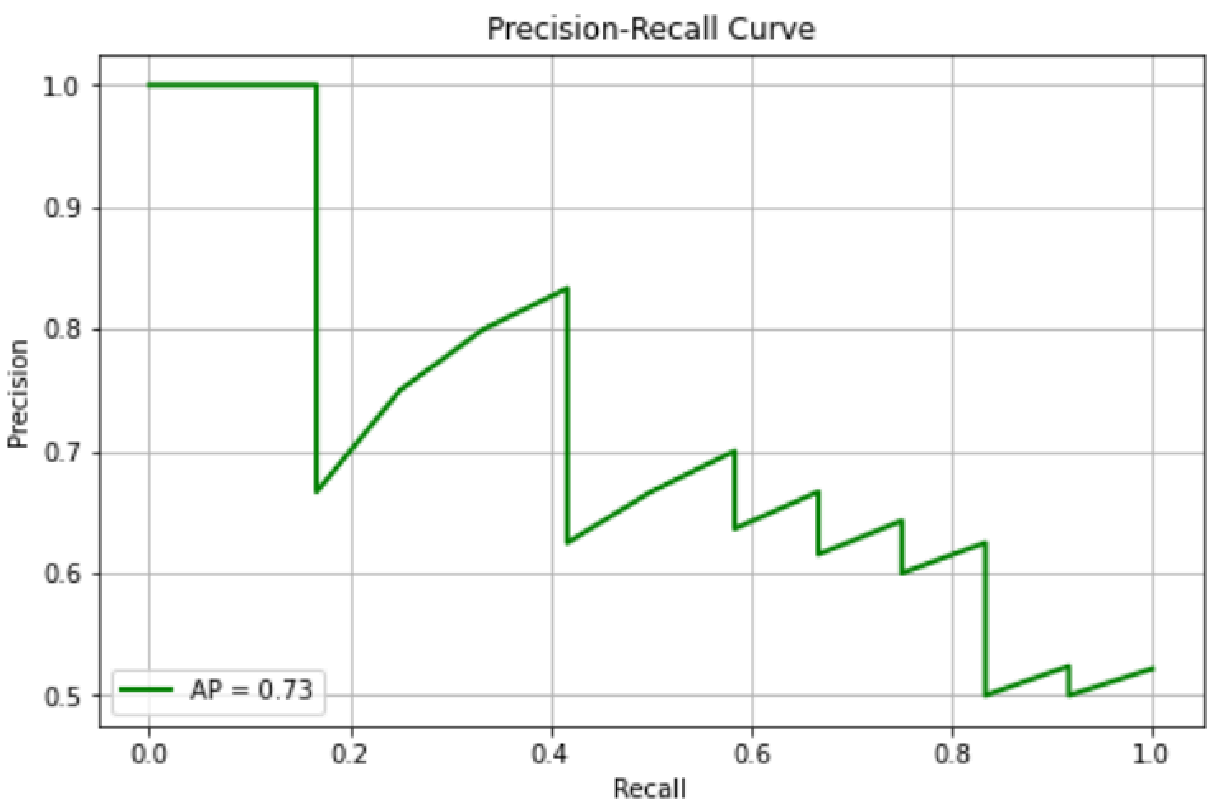
- Image augmentation:
  - Horizontal flip
  - Rotation (±10°)
  - Zoom (±10%)
  - Width and height shift (±5%)
- Batch size: 32
- Optimizer: Adam (learning rate = 1e-4)
- Loss: Binary crossentropy
- Metrics: Accuracy
- Frozen layers: All convolutional layers of VGG16
- Fine-tuned layers: None (only custom top layers trained)
- Layers added: GlobalAveragePooling2D → Dropout → Dense(128) → Dropout → Dense(1)
- Epochs: Up to 30 with early stopping (patience = 10)

Training History:
Training and validation loss and accuracy showed gradual improvement. Best model saved at epoch 20. Early stopping occurred at epoch 30 due to no further improvement in validation loss.

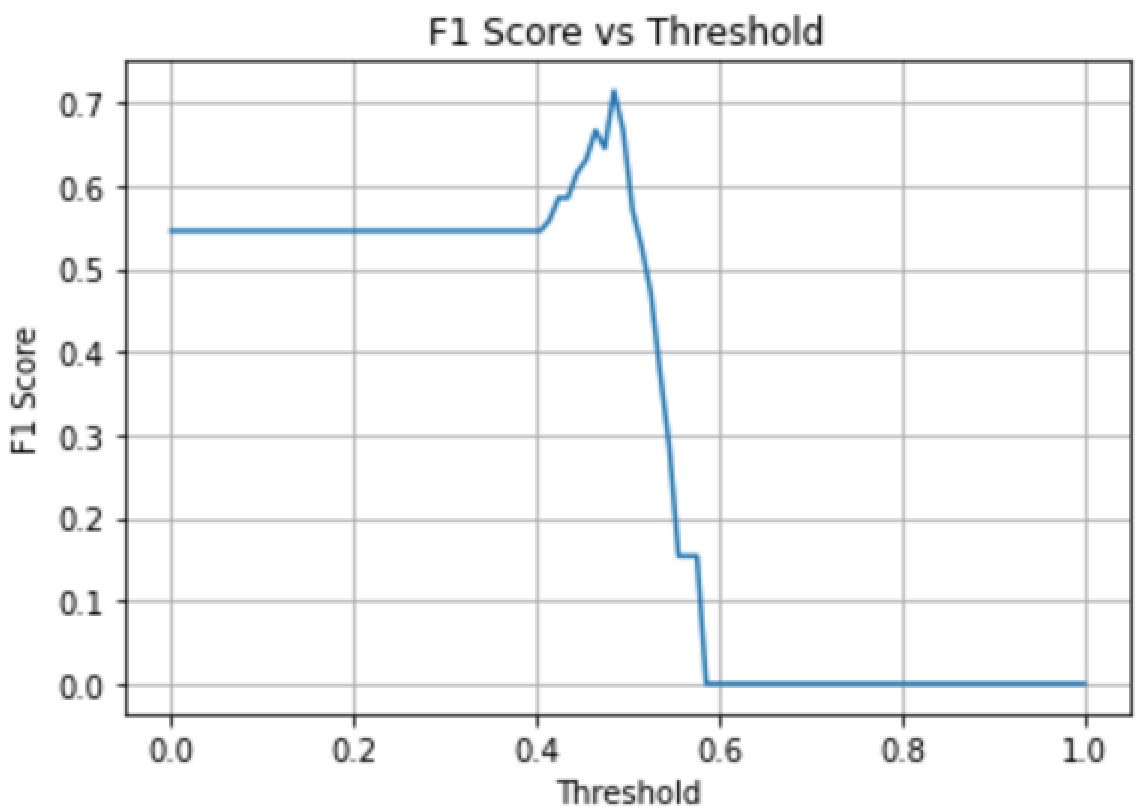
Algorithm training performance visualization



P-R curve



Final Threshold and Explanation: 0.49



A threshold of 0.49 was selected to optimize F1 score. This balance favored both precision and recall for pneumonia detection.

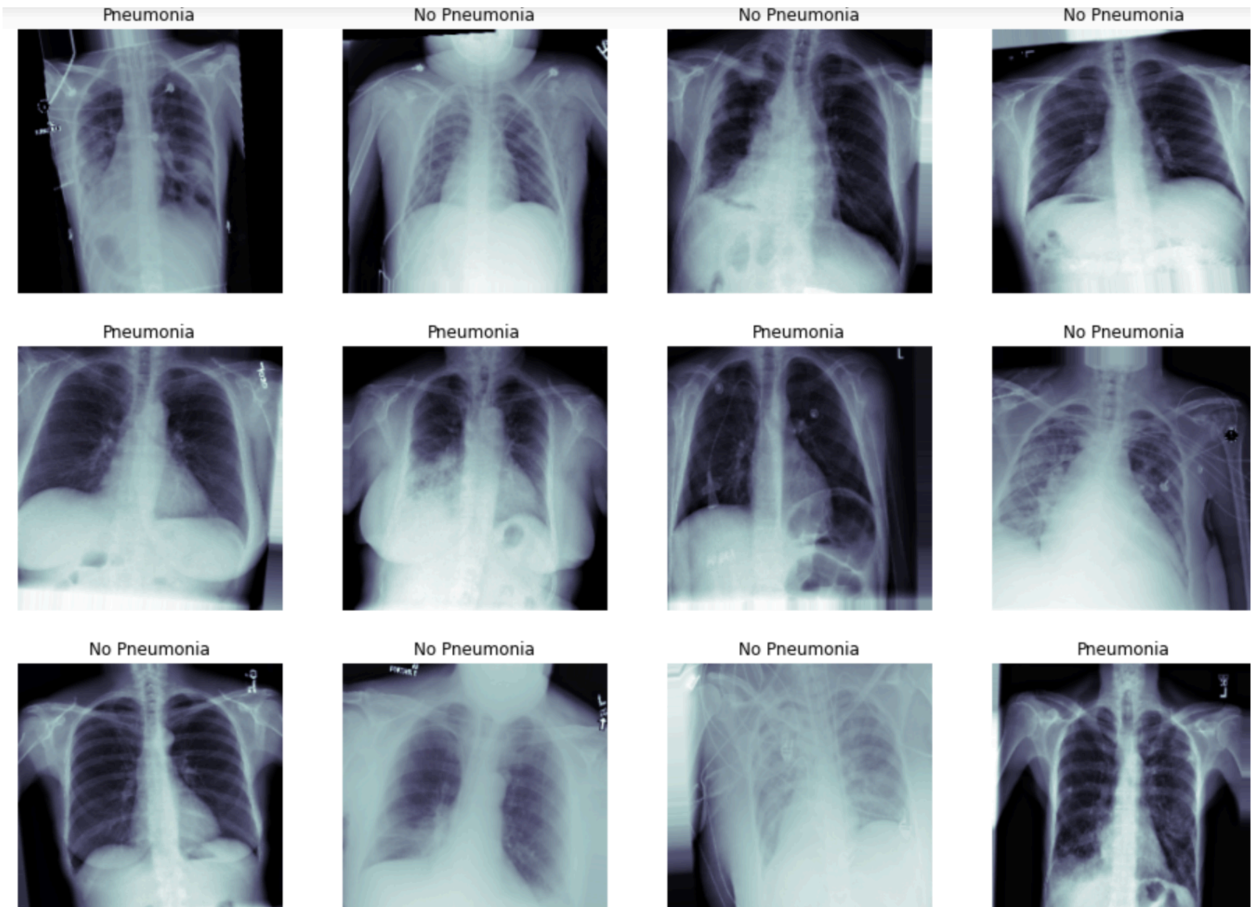
Performance at Best Threshold (Validation Set):

- Accuracy: 0.75
- Precision: 0.62 (Pneumonia)
- Recall: 0.83 (Pneumonia)
- F1 Score: 0.71 (Pneumonia)

4. Databases

Description of Training Dataset:

Visualized examples of our augmented training data



- **Source:** NIH ChestX-ray14 dataset

- The original dataset was heavily imbalanced, with over 110,000 images labeled as *No Pneumonia* and only 1,431 labeled as *Pneumonia*.

- To avoid training a biased model, all 1,431 pneumonia cases were retained, and an equal number of no-pneumonia cases (1,431) were randomly selected to create a balanced dataset.

- The final training set included **2,289 images**:

    - 1,145 *Pneumonia*

    - 1,144 *No Pneumonia*

- **Preprocessing included:**
  - Filtering for frontal-view chest X-rays (AP/PA)
  - Resizing to 224×224 pixels
  - Normalization with mean = 128.0, std = 64.0
  - Conversion to 3-channel RGB
- **Data augmentation** (training only):
  - Horizontal flip
  - ±10° rotation
  - ±10% zoom
  - ±5% width/height shift

Description of Validation Dataset:

- Drawn from the same NIH dataset, using remaining images not included in training.
- Similarly balanced to include nearly equal numbers of both classes:
  - 286 *Pneumonia*
  - 287 *No Pneumonia*
- **Total size:** 573 images
- Preprocessed identically to training data but **no augmentation** was applied.
- Used for model evaluation, early stopping, and optimal threshold selection.

5. Ground Truth

Ground truth labels were generated using natural language processing (NLP) from associated radiology reports in the NIH dataset.

These labels were used for both training and validation, although label noise is a known limitation.

6. FDA Validation Plan

Patient Population Description for FDA Validation Dataset:

The ideal FDA validation dataset would consist of adult patients aged 18 to 80 undergoing routine frontal chest X-rays (AP or PA view). The dataset should represent real-world prevalence of pneumonia (around 20–25%) and include a balanced distribution of genders. Ideally, data would be collected across multiple hospitals to capture imaging variability and population diversity. Images should exclude those with thoracic hardware, lateral views, or significant artifacts.

Ground Truth Acquisition Methodology:

Ground truth labels would be assigned through expert consensus. Each image would be reviewed by at least three board-certified radiologists. A final label would be determined by majority vote or a weighted consensus based on the radiologists' clinical experience. Radiologist agreement would be quantified using inter-rater reliability scores such as Cohen's kappa.

Algorithm Performance Standard:

The model's performance would be primarily assessed using the F1 score due to the class imbalance between pneumonia and no pneumonia cases. According to the CheXNet paper ("Rajpurkar et al., 2017"), radiologist-level performance on pneumonia detection had an average F1 score of 0.387. Our model aims to exceed this benchmark, targeting an F1 score of at least 0.4. Supporting metrics like precision, recall, and AUC (≥ 0.85) would also be reported for robustness.